The Media Connectivity Association




OVERVIEW AND UPDATES · IMMERSIVE MEDIA

# Beyond 2D Video Experiences

Evaluating video formats beyond 2D: stereoscopic, multi-view plus depth, point clouds and dynamic meshes

3GPP

OCTOBER 2026 – www.5g-mag.com

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# About the project

The documents the project rests on

## 3GPP TR 26.956

**Technical Report**

Evaluation and Characterization of Beyond 2D Video Formats and Codecs, the output of the study FS_Beyond2D.

## 3GPP TS 26.501

**and TS 26.512**

The 5G Media Streaming architecture and protocols the formats are assessed against.

## ISO/IEC 23090-5

**Standard**

Visual Volumetric Video-based Coding (V3C), referenced for evaluation.

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# Formats under evaluation

The four format families of the study, with the coding approaches evaluated for each

**Stereoscopic video** a left and right view: a fixed 3D effect, no free viewpoint

MV-HEVC | Simulcast or frame-compatible HEVC (anchor)

**Multi-view plus depth** several views with depth: a limited change of viewpoint

MIV profile of V3C | MV-HEVC

**Dense dynamic point clouds** a volumetric capture of an object or person

V-PCC | G-PCC, noted as a related technology

**Dynamic meshes** a volumetric representation with time-varying textured 3D meshes

V-DMC

The report also surveys formats under research (neural radiance fields, light fields, 3D Gaussian splatting) without evaluating them.

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# Services evaluated

The three services of the study, as listed by the reference repository

## Stereoscopic

**Live, device to device**

UE-to-UE stereoscopic video live streaming.

## Point cloud

**On demand**

Streaming of professionally produced volumetric video, a single asset containing people.

## Multi-view

**Plus depth, on demand**

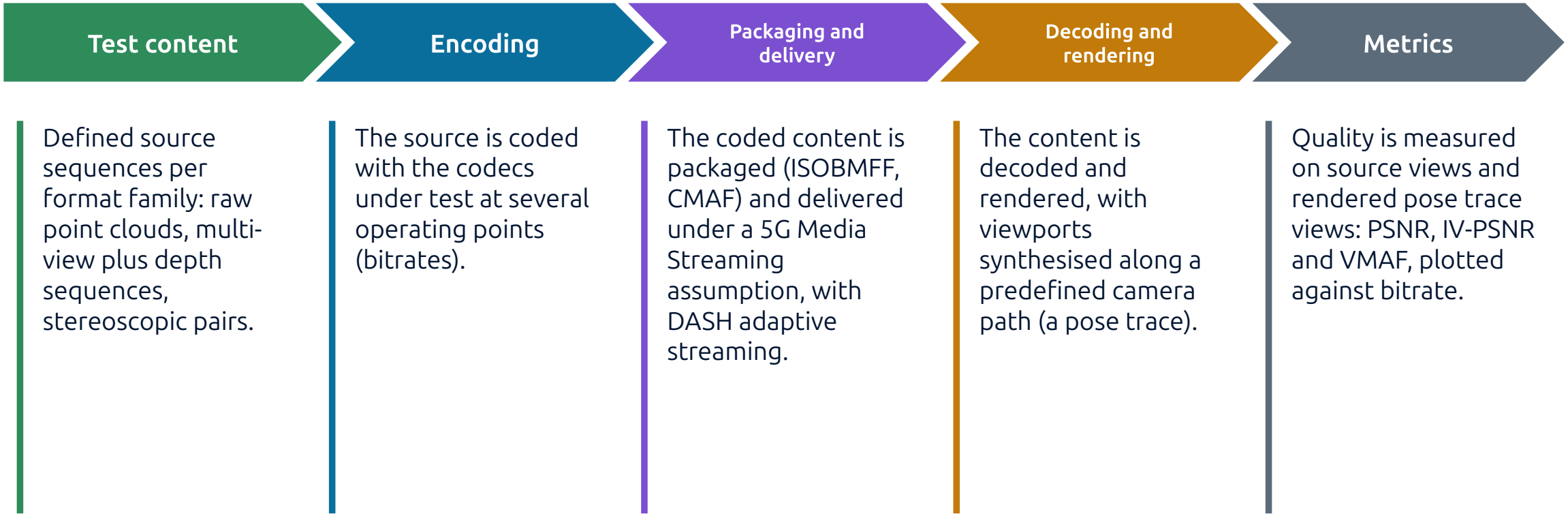
On-demand streaming of multi-view plus depth.

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# Evaluation pipeline

A repeatable pipeline, so that results are comparable across formats and contributors




Sources, bitstreams and rendered pose trace videos are large, so the framework references content on external servers.

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# How it grew

The 3GPP study and the 5G-MAG evaluation framework, from the first draft to Release 19

| April 2024 | June 2025 | June to July 2025 | September 2025 | October 2025 |
|---|---|---|---|---|
| First draft of the technical report at 3GPP SA4. | The agreed draft is sent to TSG SA for information. | First releases of the 5G-MAG evaluation framework: multi-view plus depth, then point cloud scripts. | The report is sent to TSG SA for approval; the study is complete. | TR 26.956 is published as a Release 19 Technical Report. |

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# Repository

The one repository of the evaluation framework, from its README

| REPOSITORY | WHAT IT IS | COVERS | LATEST RELEASE | |
|---|---|---|---|---|
| rt-beyond2d-evaluation-framework | scripts and configuration files for the reproducible evaluation of the FS_Beyond2D services | Multi-view plus depth · Point cloud | v0.2.1 24 Jul 2025 | GitHub |

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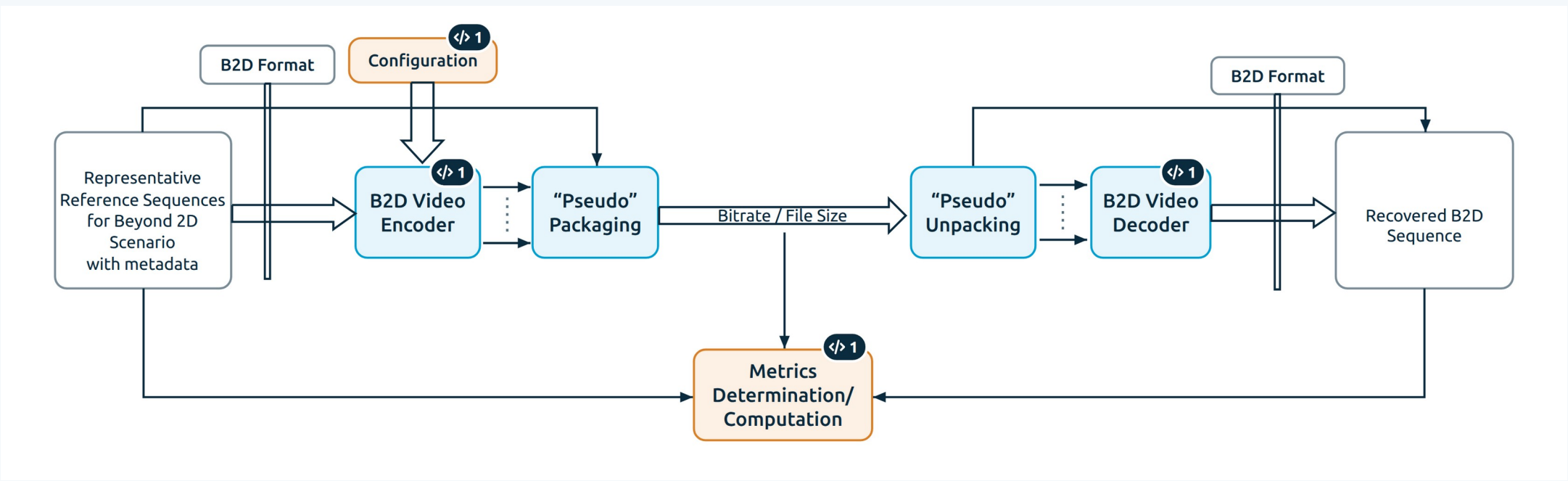
● released ○ no release yet

More information at 5g-mag.com/tech/beyond-2d

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# Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them




## PROVIDED BY THE REFERENCE TOOLS

1 **rt-beyond2d-evaluation-framework**

Mapped from the repository's MVD scenario only; boxes without a badge have no repository.

CONTRIBUTE

# Latest updates and roadmap

Status on 19 October 2026

## RELEASED

**rt-beyond2d-evaluation-framework v0.2.1**

24 Jul 2025. Adds the point cloud setup and updates for video generation scripts.

## IN DEVELOPMENT

**Evaluation services**

Completing the evaluation setups for the three services of the study.

## PLANNED

**Test content**

Collecting the initial contents for the evaluation of the FS_Beyond2D formats.

**MERGED SINCE V0.2.0** 1 pull request in 1 repository

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